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ODD END SEMESTER EXAMINATION DECEMBER- 2025

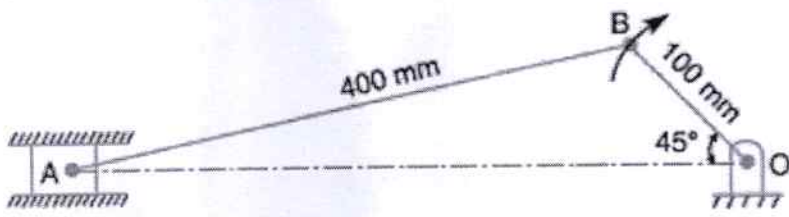
Name of the Course: **B.Tech**Semester: **V**Name of the Paper: **Theory of Machines**Paper Code: **TME 508**

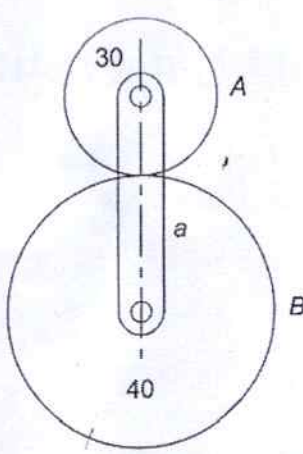
Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks.

Q1	(20marks)	
(a)	Enumerate the inversions of a double-slider-crank chain with examples.	
(b)	Define Grashof's law. State how is it helpful in classifying the four-link mechanisms into different types.	
(c)	Locate all the instantaneous centres of the slider crank mechanism as shown in figure. The lengths of crank OB and connecting rod AB are 100 mm and 400 mm respectively. If the crank rotates clockwise with an angular velocity of 10 rad/s. find: i. Velocity of the slider A ii. Angular velocity of the connecting rod AB	CO1, CO2
		
Q2	(20 marks)	
(a)	Explain the procedure to construct Klein's construction to determine the velocity and acceleration of a slider-crank mechanism.	
(b)	Give a detailed classification of gears,	
(c)	An epicyclic gear train consists of an arm and two gears A and B having 30 and 40 teeth respectively. The arm rotates about the centre of the gear A at a speed of 80 rpm counter-clockwise. Determine the speed of the gear B if: i. The gear A is fixed. ii. The gear A revolves at 240 rpm clockwise instead of being fixed.	CO2, CO3

																	
Q3	(20 marks)	CO4															
(a)	Discuss the various types of cams and followers.																
(b)	Derive an expression for the angular velocity and angular acceleration of the connecting rod.																
(c)	A horizontal gas engine running at 210 rpm has a bore of 220 mm and a stroke of 440 mm. The connecting rod is 924 mm long and the reciprocating parts weigh 20 kg. When the crank has turned through an angle of 30° from the inner dead centre, the gas pressures on the cover and the crank sides are 500 kN/m ² and 60 kN/m ² respectively. Diameter of the piston rod is 40 mm. Determine i. Turning moment on the crank shaft ii. Thrust on the bearings																
Q4	(20 marks)	CO5															
(a)	Explain the method of finding the counter masses in two planes to balance the dynamic unbalance of rotating masses.																
(b)	A rotor has the following properties: <table border="1" data-bbox="381 1313 1211 1517"><thead><tr><th>Mass</th><th>Magnitude (kg)</th><th>Radius of rotation (mm)</th><th>Angle (degree)</th></tr></thead><tbody><tr><td>1</td><td>3</td><td>30</td><td>30</td></tr><tr><td>2</td><td>4</td><td>20</td><td>120</td></tr><tr><td>3</td><td>2</td><td>25</td><td>270</td></tr></tbody></table> Angular positions are measured counterclockwise from the reference line along the x-axis. Determine the amount of counter mass at a radial distance of 35 mm required for static balance.		Mass	Magnitude (kg)	Radius of rotation (mm)	Angle (degree)	1	3	30	30	2	4	20	120	3	2	25
Mass	Magnitude (kg)	Radius of rotation (mm)	Angle (degree)														
1	3	30	30														
2	4	20	120														
3	2	25	270														
(c)	A casting weighs 6 kN and is freely suspended from a rope which makes 2.5 turns around a drum of 200 mm diameter. If the drum rotates at 40 rpm, determine the force required by a man to pull the rope from the other end of the rope. Also find the power to raise the casting. The coefficient of friction is 0.25.																
Q5	(20 marks)	CO6															
(a)	Explain the working of Watt Governor with a neat sketch.																
(b)	What do you mean by gyroscopic couple? Derive a relation for its magnitude.																
(c)	A porter governor has each of its arms of 175 mm length pivoted on the axis of the governor. The radii of rotation of the balls at minimum and maximum speeds are 105 mm and 140 mm respectively. The mass of the sleeve is 20 kg and of each ball is 5 kg. Find the range of speed of the governor. Also determine the range of speed when the friction at the sleeve is 15 N.																